0.1 Setting up the ODE for the Numerical Solution

0.1.1 ODE for the Flow of the Adimensionless Potential

One starts by considering the flow equation at the phase transition, Eq. (0.1).

$$0 = -d\tilde{U}_t + (d - s + \eta)\,\tilde{\rho}\tilde{U}_t^{(1)} + \frac{A_d \frac{s}{2}\left(1 - \frac{\eta}{s+d}\right)}{1 + \tilde{U}_t^{(1)} + 2\tilde{\rho}\tilde{U}_t^{(2)}} \tag{0.1}$$

The prefactors in the last term can be removed by using the identities, Eq. (0.2).

$$\tilde{\rho}' = \xi\tilde{\rho}\ ; \qquad \tilde{U}_t' = \xi\tilde{U}_t\ ; \qquad \xi^{-1} = A_d\frac{s}{2}\left(1 - \frac{\eta}{s+d}\right) \tag{0.2}$$

Namely, one notices Eqs. (0.3)

$$\frac{\partial\tilde{U}_t}{\partial\tilde{\rho}} = \xi^{-1}\frac{\partial\tilde{\rho}'}{\partial\tilde{\rho}}\frac{\partial\tilde{U}_t'}{\partial\tilde{\rho}'} = \frac{\partial\tilde{U}_t'}{\partial\tilde{\rho}'}\ ; \qquad \frac{\partial^2\tilde{U}_t}{\partial\tilde{\rho}^2} = \frac{\partial\tilde{\rho}'}{\partial\tilde{\rho}}\frac{\partial^2\tilde{U}_t'}{\partial\tilde{\rho}'^2} = \xi\frac{\partial^2\tilde{U}_t'}{\partial\tilde{\rho}'^2} \tag{0.3}$$

Which reduce Eq. (0.1) onto Eq. (0.4).

$$0 = -d\tilde{U}_t + (d - s + \eta)\,\tilde{\rho}\tilde{U}_t^{(1)} + \frac{1}{1 + \tilde{U}_t^{(1)} + 2\tilde{\rho}\tilde{U}_t^{(2)}} \tag{0.4}$$

By inverting Eq. (0.4), one obtains Eq. (0.5).

$$\tilde{U}_t^{(2)} = -\frac{1}{2\tilde{\rho}}\left(1 + \tilde{U}_t^{(1)} + \frac{1}{-d\tilde{U}_t + (d - s + \eta)\,\tilde{\rho}\tilde{U}_t^{(1)}}\right) \tag{0.5}$$

Satisfying the initial conditions, Eq. (0.6).

$$\tilde{U}_t^{(1)}(\tilde{\rho} = 0) = \sigma\ ; \qquad \tilde{U}_t(\tilde{\rho} = 0) = \frac{1}{d\,(1 + \sigma)} \tag{0.6}$$

In practice, this equation is, however, not very convenient for numerical integrations. This is because the denominator inside the parenthesis in Eq. (0.5) introduces a non-physical boundary when the denominator tends to zero, which may lead to numerical instabilities. By instead diferentiating Eq. (0.1) with respect to $\tilde{\rho}$, one obtains Eq. (0.7).

$$0 = (-s+\eta)\,\tilde{U}_t^{(1)} + (d-s+\eta)\,\tilde{\rho}\tilde{U}_t^{(2)} - \frac{3\tilde{U}_t^{(2)} + 2\tilde{\rho}\tilde{U}_t^{(3)}}{\left(1+\tilde{U}_t^{(1)} + 2\tilde{\rho}\tilde{U}_t^{(2)}\right)^2} \tag{0.7}$$

That is, Eq. (0.8).

$$\tilde{U}_t^{(3)} = \frac{1}{2\tilde{\rho}}\left\{\left(1+\tilde{U}_t^{(1)} + 2\tilde{\rho}\tilde{U}_t^{(2)}\right)^2\left[(-s+\eta)\,\tilde{U}_t^{(1)} + (d-s+\eta)\,\tilde{\rho}\tilde{U}_t^{(2)}\right] - 3\tilde{U}_t^{(2)}\right\} \tag{0.8}$$

The initial conditions are correspondingly given by Eq. (0.9).

$$\tilde{U}_t^{(1)}\,(\tilde{\rho}=0) = \sigma\ ; \qquad\qquad \tilde{U}_t^{(2)}\,(\tilde{\rho}=0) = \frac{(-s+\eta)\,(1+\sigma)^2\,\sigma}{3} \tag{0.9}$$

Eventually, Eq. (0.8) reaches an asymptotic state, where the denominator term goes to infinity and the third term vanishes. In this case, the equation can be solved analytically to Eq. (0.10), where $\tilde{\rho}_\infty$ is the field from which the asymptotic equation becomes valid,and $\tilde{\rho} > \tilde{\rho}_\infty$.

$$\tilde{U}_t\,(\tilde{\rho}) = \tilde{U}_t\,(\tilde{\rho}_\infty)\left(\frac{\tilde{\rho}}{\tilde{\rho}_\infty}\right)^{\frac{d}{d-s+\eta}} \tag{0.10}$$

Finally, by remembering that $\tilde{\rho} = \frac{1}{2}\tilde{\phi}^2$, it is possible to instead consider solutions in terms of Eq. (0.11). The solutions in this case are simiar, except that the initial condition is affixated by $\tilde{U}_t^{(2)}\left(\tilde{\phi}=0\right) = \sigma$ instead.

$$0 = -d\tilde{U}_t + \frac{(d-s+\eta)}{2}\,\tilde{\phi}\tilde{U}_t^{(1)} + \frac{A_d\frac{s}{2}\left(1-\frac{\eta}{s+d}\right)}{1+\tilde{U}_t^{(2)}} \tag{0.11}$$

Out of the possibilities above, we noticed best numerical stability by using Eq. (0.8) and the initial condition, Eq. (0.9). Furthermore, the identification of a threshold $\tilde{\rho}_\infty$ and the replacement of the numerical solution by the analytical solution, Eq. (0.10), for all $\tilde{\rho} > \tilde{\rho}_\infty$, led to significantly improved numerical stability, particularly for the calculation of the RG eigenvalues.

0.1.2 ODE for the RG Eigenvectors

One considers an infinitesimal perturbation in the vicinity of the fixed point, Eq. (0.11), with $\varepsilon \to 0$.

$$\tilde{U}_t = \varepsilon\tilde{\nu}\,(y,\tilde{\rho})\,e^{-yt} \tag{0.12}$$

By substituting Eq. (0.12) onto Eq. (0.8) and taylor expanding the denominator to the first order in

ε,one obtains Eqs. (0.13, 0.14).

$$0 = -\left(-y+d\right)\tilde{\nu} + \left(d-s+\eta\right)\tilde{\rho}\tilde{\nu}^{(1)} - \frac{\tilde{\nu}^{(1)} + 2\tilde{\rho}\tilde{\nu}^{(2)}}{\left(1+\tilde{U}_t^{(1)} + 2\tilde{\rho}\tilde{U}_t^{(2)}\right)^2} \tag{0.13}$$

$$\tilde{\nu}^{(2)} = -\frac{1}{2\tilde{\rho}}\left\{\left(1+\tilde{U}_t^{(1)} + 2\tilde{\rho}\tilde{U}_t^{(2)}\right)^2\left[\left(-y+d\right)\tilde{\nu} - \left(d-s+\eta\right)\tilde{\rho}\tilde{\nu}^{(1)}\right] + \tilde{\nu}^{(1)}\right\} \tag{0.14}$$

Since $\tilde{\nu}$ is a degree of freedom, one may choose the initial conditions, Eq. (0.15).

$$\tilde{\nu}\left(\tilde{\rho}=0\right) = 1\ ; \qquad \tilde{\nu}^{(1)}\left(\tilde{\rho}=0\right) = \left(1+\sigma\right)^2\left(d-y\right) \tag{0.15}$$

Eq. (0.14) also has an asymptotic solution, Eq. (0.16). Although this is not directly used for the purposes of the numerical integration described in this paper, it could in theory lead to an improvement in numerical results.

$$\tilde{\nu}\left(\tilde{\rho}\right) = \tilde{\nu}\left(\tilde{\rho}_\infty\right)\left(\frac{\tilde{\rho}}{\tilde{\rho}_\infty}\right)^{\frac{-y+d}{d-s+\eta}} \tag{0.16}$$

0.2 Simulated Annealing

As a first step, one must find the parameters (σ, η) for which the potential, Eq. (), is (or at the very least appears to be) analytical. Those results correspond to possible phase transitions that would therefore warrant further investigation. For cases where the anomalous dimension is already known, a simple grid search over possible values of σ appears to be an efficient solution. On the other hand, when η is also unknown, the following main problems emerge:

1. η is not a free variable, but it is rather dependant on σ,that is to say, $\eta = \eta(\sigma)$. This means that, in practice, most of the solutions for (σ, η), were η to be treated as an independent variable, are actually spurious solutions and do not represent actual, physical phase transitions.

2. It appears that a possible solution would be to choose some values (σ, η), integrate the potential, and self-consistently re-calculate η based on the numerically-integrated trajectory for the potential. On the other hand, unless (σ, η) is already sufficiently close to a physical solution, there is no guarantee that a re-caluclation of η will at all ensure that the system converges. Furthemore, if more than a single meta-stable solution exists, one runs onto the risk of failing to capture all possible solutions.

3. Finally, a grid search over all parameters (σ, η) is generally unrealistic due to the computational overhead. In practice, a more efficient method is required, at least in order to detect the *general region* where the solution-space lays, such that a grid search can then be used within this region only.

4. Even if a reduced solution-space can be found, as described in (3) above, one is still not guaranteed to find good solutions with a grid search, because the solution space may be (in fact, it is) significantly stiff.

From numerical simulations realized in connection with this thesis, a reliable solution that addresses the issues described above appears to consist of implementing a monte-carlo-simulated-annealing algoritm to conduct the grid search. Essentially, one defines a "temperature" variable that decays according toa cooling rate, Eq. (0.17). For each simulation:

$$T = -\kappa T dt \tag{0.17}$$

1. Choose (σ, η) at random.

2. Integrate the potential $\tilde{U}_t(\tilde{\rho}, \sigma, \eta)$, find the threshold $\tilde{\rho}_\infty^{(\sigma,\eta)}$ for which the numerical solution diverges. Furthermore, potentially re-calculate η_{new} self-consistently, given the calcualted trajectory $\tilde{U}_t(\tilde{\rho}, \sigma, \eta)$. Potentially recalculate η_{new} as many times as necessary to ensure convergence or up until a maximum number of iterations.

3. Compare with previous iteration, Eq. (), where the index i indicates the iteration number.

4. Accept all configurations that improve the threshold. If threshold is worsened, accept or reject the new configuration based on an exponental probability $P \sim e^{-\Delta_i T}$, where T is the temperature.

5. Update the temperature, based on the cooling rate, by means of Eq .(0.17).

$$\Delta_i = -\frac{\tilde{\rho}_{\infty,i}^{(\sigma,\eta)} - \tilde{\rho}_{\infty,i-1}^{(\sigma,\eta)}}{\tilde{\rho}_{\infty,i}^{(\sigma,\eta)}} \tag{0.18}$$

As a general strategy, one begins with a strong cooling rate, $\kappa >> 0$, and a small number of iterations, $i_{max} << \infty$. The algorithm is re-initialized at multiple values of (σ, η). This ensures that, at first instance, the algorithm will explore a broad region of the solution space and, accordingly, identify many local maxima for the asymptotic field, that is, many regions where phase transitions may occur. Once those portions are identified, a second annealing phase is executed, in order to properly characterize those regions of local maxima, and to identify the exact point corresponding to the phase transtion in such vicinity. For this second annealing phase, a much smaller cooling rate is chosen, $\kappa \to 0$, and respectively $i_{max} >> 0$. This ensures that the correct maxima is identified. A subsequent grid search in the "infinitesimal" region near the located maxima may be executed, to guarantee even further convergence. If this approach is chosen, it is important to ensure (1) that the point itself where the annealing predicted the maxima belongs to the grid; and (2) that the search window gets subsequently "zoomed-in" in order to improve decimal point accuracy.

An important remark to emphasize is that, whilst in the first annealing phase, the value of the anomalous dimension is fixed by σ (that is, the result is self-consistently re-calculated), in the second annealing phase, one allows η to be a free-floating parameter. This is justified on the basis that (1) within an attraction basin, the maxima is clearly distinguishable by a unique choice of (σ, η) (this statement is based on the visual observation of the graphs); and (2) therefore it makes sense to let the anomalous dimension float freely in this vicinity, in order to remove the effect of numerical artifacts induced to the anomalous dimension calculation by the numerical integration of the potential.

0.3 Effective dimension approach

It appears that, to first order, long-ranged models can be related to respective short-ranged models by means of the transformation, Eq. (0.19).

$$D_{eff} = \frac{d}{s}\left(2 - \eta\left(D_{eff}\right)\right) \tag{0.19}$$

Namely, starting from Eq. (0.20), one obtains Eq. (0.20).

$$0 = -D_{eff}\tilde{U}_t + \left(D_{eff} - 2 + \eta\right)\tilde{\rho}\tilde{U}_t^{(1)} + \left(1 - \frac{\eta}{D_{eff}+2}\right)\left(1 + \tilde{U}_t^{(1)} + 2\tilde{\rho}\tilde{U}_t^{(2)}\right)^{-1} \tag{0.20}$$

$$0 = -d\tilde{U}_t + (2 - s)\,\tilde{\rho}\tilde{U}_t^{(1)} + \frac{s}{2}\left(\frac{1}{1-\frac{\eta}{2}}\right)\left(1 - \frac{\eta}{D_{eff}+2}\right)\left(1 + \tilde{U}_t^{(1)} + 2\tilde{\rho}\tilde{U}_t^{(2)}\right)^{-1} \tag{0.21}$$

In the least extreme case, $D_{eff} \to \infty$ when $s \to 0$. In this case, $\eta\left(D_{eff}\right) \to 0$ since the anomalous dimension tends to drop with dimensionality, being always zero for $d \geq 4$ (at least for scalar theories). But when $\eta = 0$, the residual terms inside the parenthesis become exactly equal to one, Eq. (0.22).

$$\left(\frac{1}{1-\frac{\eta}{2}}\right)\left(1 - \frac{\eta}{D_{eff}+2}\right) = 1 \tag{0.22}$$

The effective dimension approach is therefore <u>exact</u> for $d \geq 4$, and it is expected to reproduce fairly accurate results for $d \lesssim 4$, since in this limit η is small and the residual terms vanish almost. On the other hand, the most extreme case occurs when $D_{eff} \to d$. In this case, for the bi-critical model in $d = 2$, which (from CFT) one expects to dispose of the greatest anomalous dimension, $\eta = 0.25$, one obtains $s = 2 - \eta$ and the residual term, Eq. (0.23). In other words, $D_{eff} \to d$ corresponds to the Sak Point $s^{\star} = 2 - \eta$, and in this limit the residual coefficients in the third term of Eq. (0.21) report an error of around 7% in relation to the long-ranged equation.

$$\left(\frac{1}{1-\frac{\eta}{2}}\right)\left(1-\frac{\eta}{D_{eff}+2}\right) \approx 1.07 \tag{0.23}$$

Because the Ansatz truncation lacks a characteristic stopping point,

0.4 Simulation Configurations

Due to computational limitations that arise in the context of numerical computation, artifacts must be introduced to the numerical implementation, in order to ensure stability.

- The asymptotic field is integrated to, at most, $\tilde{\rho}_{\infty} = 2000$ units. If a trajectory is integrated up to this value, the trajectory is treated as if it had converged.

- The asymptotic field is integrated starting from $\tilde{\rho}_0 = 10^{-6}$ units (as opposed to $\tilde{\rho}_0 = 0$ units), and it is integrated in "inifinitesimal" of $\Delta\tilde{\rho} = 10^{-6}$ using a `runge_kutta_fehlberg78` integrator from the C++ library `boost/numeric/odeint.hpp`. Both the tolerance and the absolute tolerances are set to 10^{-10}, and the algorithm implements adaptative steps, such that, if the convergence fails, step sizes are reduced by 0.5 units per iteration, up to $\Delta\tilde{\rho}_{max} = 10^{-8}$ units. The grid is afterwards relaxed by a Hermite interpolation algorithm, such that a homogenous grid with $8'000$ steps is ultimately sampled.

- For the cases where the anomalous dimension is self-consistently recalculated, 100 re-calculations at most are allowed, with a convergence criteria $|\eta_{old} - \eta_{new}| \leq 10^{-4}$. Furthermore, the anomalous dimension is updated by Eq. (0.24), where the damping is set to $\delta = 0.05$.

$$\eta_{new} = \delta \times \eta_{old} + (1-\delta) \times \eta_{recalculated} \tag{0.24}$$

- For cases where a grid search is implemented, the grid search is repeated for at most 5 recalculations and, at each recalculation, the interval is zoomed-in by a factor of 4 and subdivided in 101 points (an odd number is chosen to ensure that the best previous estimate belongs to the grid).

- For the cases where an asymptotic patching is implemented, $\tilde{U}_t$ and its derivatives integrate to at most $\tilde{U}_{t,max} = 10'000$ units.

- For runs in connection to the shooting method or the grid search method, normally initial conditions ranging from $\sigma_{min} = -0.95$ to $\sigma_{max} = +0.95$ in $8'000$ steps are used, although at times σ_{max} is extended up to $\sigma_{max} = 2.50$ and the numer of steps is extended to at most $8'000$ steps, if it is observed that the chosen interval doesn't encompass all the relevant multicritical points. For example, for $d = 2.05$ and $s = 2$, at least the tri-critical point lies between $\sigma \in [2.00, 2.50]$, in which case additional sampling is required.

- For runs in connection to the "eigenperturbation method" (that is, the method for finding the RG eigenvalues by means of Eq. (0.14)), we consider eigenvalues $y \in [-3.00, +3.00]$ to at most $2'000$ steps. The initial condition σ and the anomalous dimension η are chosen from the shooting method, and a subsequent grid search method in the vicinity of the input parameter (to a range of $\pm 10\%$ of the input parameter) is executed to tune the values of σ, η, in order to improve convergence. Furthermore, in this case η is treated as an independent variable as opposed to be calculated self-consistently from σ, as already discussed previusly.

0.5 Example Results

In order to exemplify the methodology described above, we display here some example results. Figure 0.1 shows, on the left, a plot corresponding to the shooting method for $D_{eff} = 2.40$, whereas the plot on the right for $D_{eff} = 2.05$. Each of the spikes corresponds to a possible multicritical point (Notice, for example, the right-most spike in the plot for $D_{eff} = 2.40$: this is a spurious spike, as we discuss in mode details later). Similarly, also for exemplification reasons, we display in Figure ... the plot of the eigenvector $\tilde{\nu}_{\infty}^{(y)}$ as function of the trial eigenvalue y, on the left for the long-range hexacritical points for $d = 2$, $s \in \{1.85, 1.90, 1.95\}$, and on the right for the tetracritical model at different effective dimensions D_{eff}.

0.6 Anomalous Dimensions and Sak Points for Multicritical Models

As a first step, we would like to calculate the anomalous dimensions and corresponding Sak points for the first five multicriticalmodels. We choose fractional dimensions $D_{eff} \in \{3.00, 2.80, 2.60, 2.40, 2.20, 2.10, 2.05, 2.01, 2.001\}$, and we execute the shooting method with self-consistent re-calculation of the anomalous dimension. Because convergence is not guaranteed, in practice we try multiple initial conditions for the anomalous dimensions, $\eta_{start} \in \{0.05, 0.10, 0.15, 0.25\}$. Eq. (0.19) can then subsequently be used to determine the long-ranged model corresponding to the short-ranged model in the effective dimension. In Tables 1 and 2, we display, respectively, the anomalous dimensions and the initial conditions at which particular multicritical points in particular effective dimensions occur, with the row correpsonding to $D_{eff} = 2.001$ corresponding, roughly, to the anomalous dimension of the correpsonding CFT model. Table 3, on the other hand, display the corresponding long-ranged exponent s corresponding to a particular multicritical poin tin a particular effective

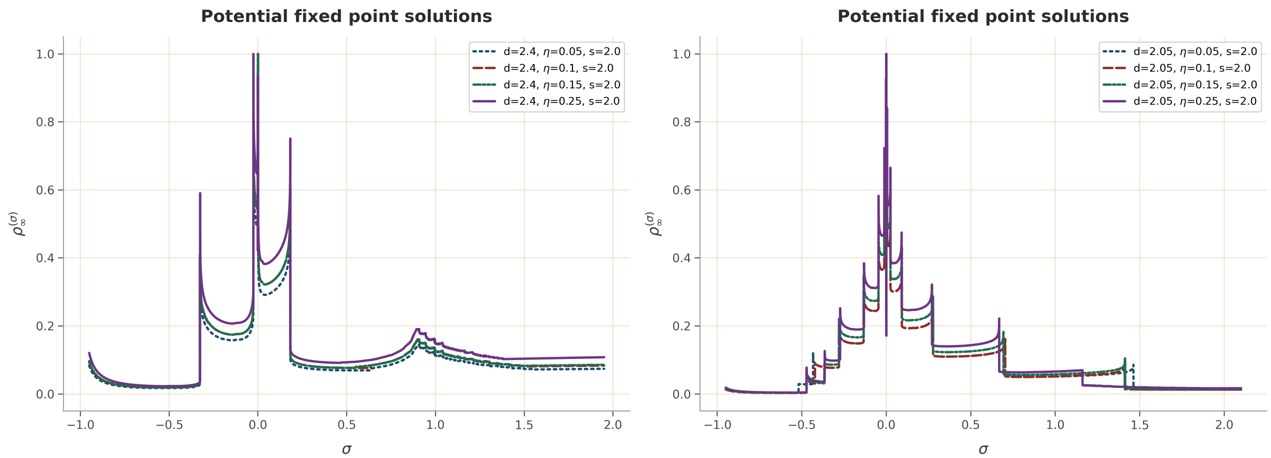

Figure 0.1: An exemplification of the shooting method, applied to $D_{eff} = 2.40$ and $D_{eff} = 2.05$, respectively for the left-most and right-most plot. The anomalous dimension in this case is calculated self-consistently, and the labels indicating lines for $\eta \in \{0.05, 0.10, 0.15, 0.25\}$ correspond to the initial condition η_{start} for which this line was generated.

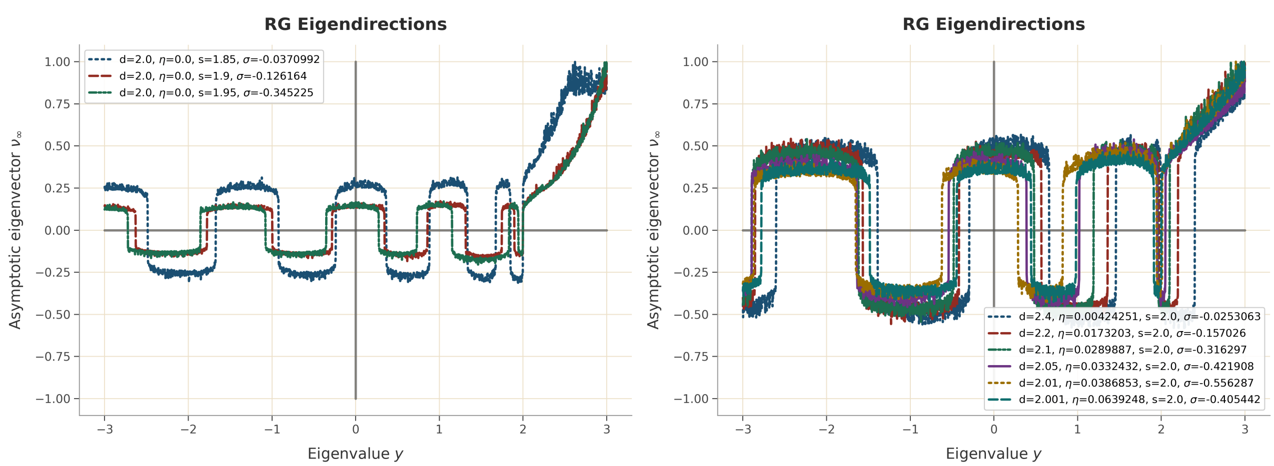

Figure 0.2: Plots for the RG eigendirections.

Dimension	Bicritical	Trictricial	Tetracritical	Pentacritical	Hexacritical
3	0.0587				
2.8	0.0900				
2.6	0.1279	0.0084			
2.4	0.1715	0.0238	0.0042		
2.2	0.2190	0.0500	0.0173	0.0070	
2.1	0.2430	0.0651	0.0290	0.0151	0.0088
2.05	0.2555	0.0730	0.0332	0.0201	0.0129
2.01	0.2646	0.0812	0.0387	0.0209	0.0150
2.001	**0.2677**	**0.0821**	**0.0639**	**0.0388**	**0.0211**

Table 1: The Anomalous Dimension for different effective dimensions ranging from $D_{eff} = 3$ to $D_{eff} = 2.001$, for the first 5 multi-critical points. Notice that the last row corresponds, roughly, to the anomalous dimensions of the respective multicritical points, on the basis that the Sak points occur when $D_{eff} \to 2$.

Dimension	Bicritical	Trictricial	Tetracritical	Pentacritical	Hexacritical
3	-0.159				
2.8	-0.208				
2.6	-0.263	0.061			
2.4	-0.327	0.183	-0.025		
2.2	-0.403	0.527	-0.157	0.061	0.000
2.1	-0.448	0.991	-0.316	0.291	-0.116
2.05	-0.472	1.463	-0.422	0.704	-0.280
2.01	-0.494	2.095	-0.556	1.566	-0.492
2.001	-0.498	2.335	-0.405	1.445	-0.401

Table 2: The respective initial conditions $\sigma_i^{D_{eff}}$ at which, for effective dimensions ranging from $D_{eff} = 3$ to $D_{eff} = 2.001$, for the first 5 multi-critical points were found.

dimension, as calculated from Eq. (0.19); and we notice that the last row in Table 3 corresponds, roughly to the Sak points of the respective universality classes. The anomalous dimensions for the different n-critical models, as function of the effective dimension, are furthermore displayed in Figure 0.3.

The respective RG eigenvalues for the first five multicritical models, in effective dimension, are calculated with the shooting method, from Eqs. (0.14, 0.15). The results are displayed in Table 4 and Figure 0.4. Figure 0.5 displays the same as Figure 0.4, except by converting the effective dimensions of the short-range models to the respective long-range exponent s corresponding to the respective effective dimension.

0.7 Long-Ranged Model

A similar analysis is carried out for the long-ranged model, starting from Eq. (0.8) and Eqs. (0.14, 0.15), by setting $1.0 < s < 2.0$ and $\eta = 0.0$. Figure 0.6 displays the emergence of new multicritical points as function of the long-range exponent s. The results are equivalently displayed in Table 5. The respective RG eigenvalues for the first eight long-raged models are displayed in Figure 0.7 and shown in Table 6.

Dimension	Bicritical	Trictricial	Tetracritical	Pentacritical	Hexa
3	1.294				
2.8	1.364				
2.6	1.440	1.532			
2.4	1.524	1.647	1.663		
2.2	1.619	1.773	1.802	1.812	
2.1	1.673	1.843	1.877	1.890	1.896
2.05	1.702	1.880	1.919	1.932	1.939
2.01	1.727	1.909	1.952	1.969	1.975
2.001	**1.731**	**1.917**	**1.935**	**1.960**	**1.978**

Table 3: The s factor corresponding to the respective long-ranged model associated with the short-ranged model in the effective dimension. Notice that the last column corresponds, roughly, to the Sak point of the respective multicritical point, calculated from Eq. (0.19) on the basis that the Sak point occurs when $D_{eff} \to 2$.

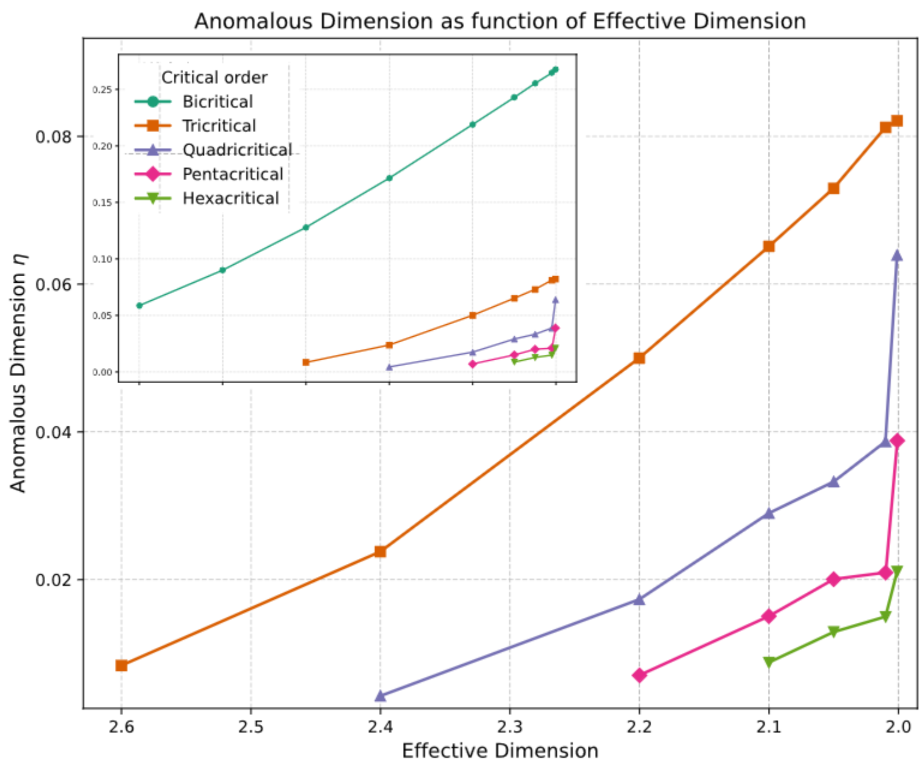

Figure 0.3: The anomalous dimensions for the first five multicritical models, as function of the anomalous dimension. In the inset, the zoomed-out version of the graph, displaying also the result for the bicritical model, whose anomalous dimension lays significantly above the other models.

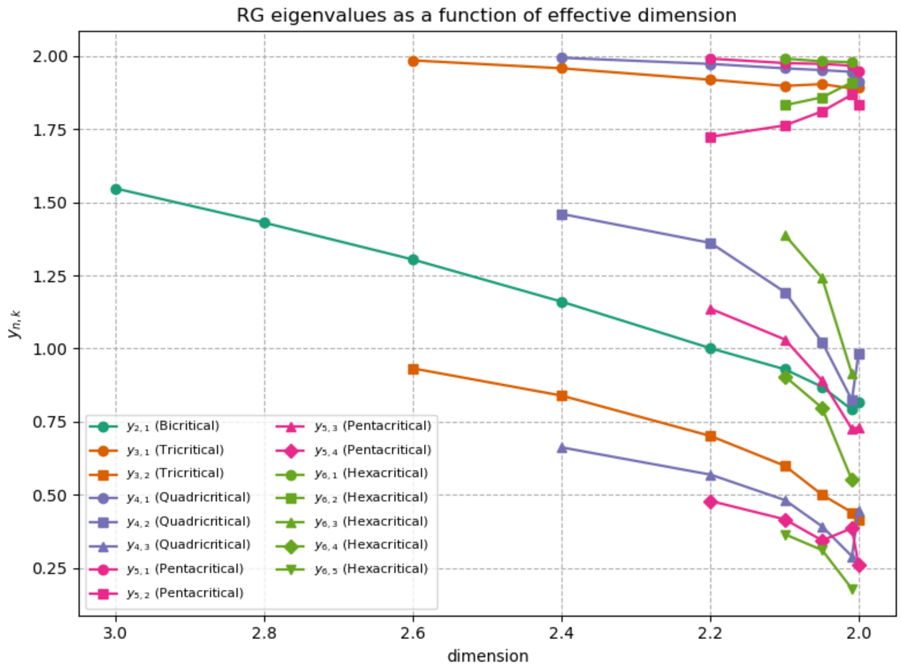

Figure 0.4: The RG eigenvalues as function of the effective dimension, for the eigenvalues corresponding to the first five multicritical points.

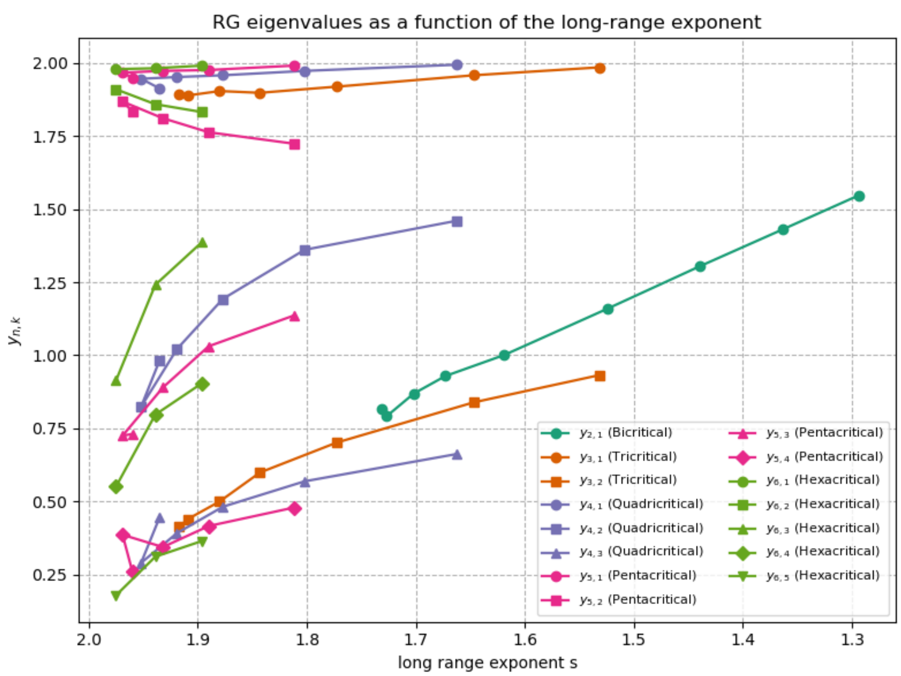

Figure 0.5: The RG eigenvalues as a function of the long range exponent s for the first five CFT multicritical models, up to the respective Sak Points, all factors determined from the effective dimension approach.

dimension	**3**	**2.8**	**2.6**	**2.4**	**2.2**	**2.1**	**2.05**	**2.01**	**2.001**
$y_{2,1}$	1.547	1.430	1.304	1.160	1.001	0.929	0.869	0.791	0.815
$y_{3,1}$			1.985	1.958	1.919	1.898	1.904	1.889	1.892
$y_{3,2}$			0.932	0.839	0.701	0.599	0.500	0.440	0.413
$y_{4,1}$				1.995	1.973	1.958	1.952	1.946	1.913
$y_{4,2}$				1.460	1.361	1.193	1.022	0.824	0.983
$y_{4,3}$				0.662	0.569	0.482	0.392	0.290	0.446
$y_{5,1}$					1.992	1.976	1.973	1.967	1.949
$y_{5,2}$					1.724	1.763	1.811	1.868	1.835
$y_{5,3}$					1.136	1.031	0.890	0.725	0.731
$y_{5,4}$					0.479	0.416	0.344	0.386	0.260
$y_{6,1}$						1.992	1.982	1.979	
$y_{6,2}$						1.832	1.859	1.910	
$y_{6,3}$						1.388	1.241	0.914	
$y_{6,4}$						0.905	0.797	0.551	
$y_{6,5}$						0.365	0.311	0.179	

Table 4: The RG eigenvalues for the first five multicritical models, for the respective effective dimensions discussed in this particular section of the thesis.

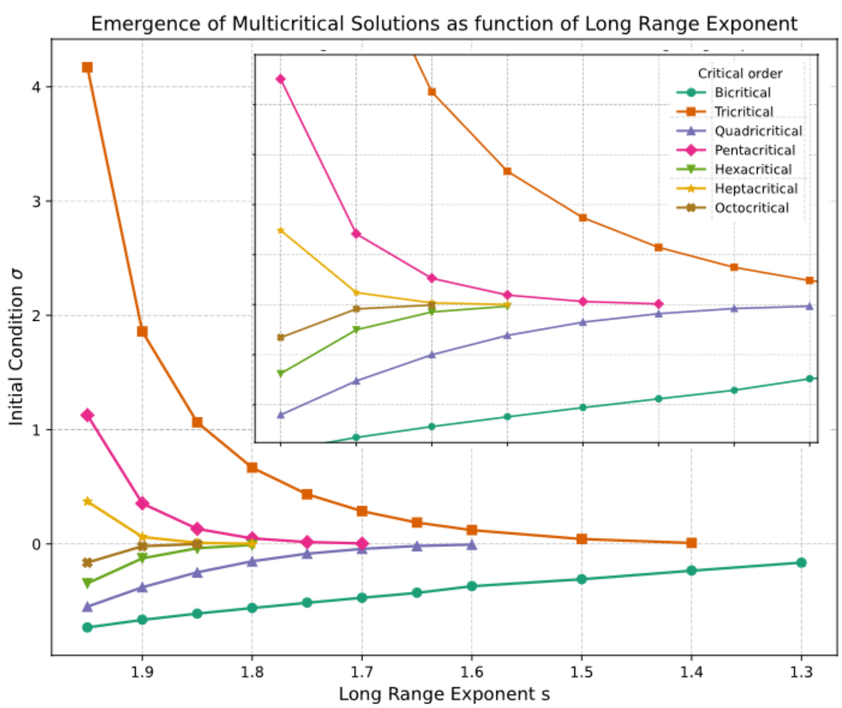

Figure 0.6: The emergence of multicritical points as a function of the long range exponent s. The y-axis displays the initial condition $\tilde{U}_t^{(1)}(\tilde{\rho} = 0) = \sigma$ that triggers the RG flow towards the particular multicritical point, whereas the x-axis displays the particular long-range exponent at which this solution σ exists.

s Factor	Bicritical	Tricritical	Tetracritical	Pentacritical	Hexacritical	Heptacritical	Octocritical
1.3	-0.163						
1.4	-0.234	0.008					
1.5	-0.309	0.042					
1.6	-0.370	0.121	-0.007				
1.65	-0.428	0.187	-0.019				
1.7	-0.470	0.287	-0.045				
1.75	-0.514	0.435	-0.087	0.016			
1.8	-0.560	0.667	-0.153	0.048	0.004		
1.85	-0.609	1.064	-0.250	0.132	-0.037	-0.001	
1.9	-0.664	1.858	-0.381	0.354	-0.126	0.060	-0.021
1.95	-0.731	4.172	-0.550	1.127	-0.345	0.372	-0.164

Table 5: The respective initial condition σ for which particular multicritical points were found, given a multicritical CFT and a long-range exponent s.

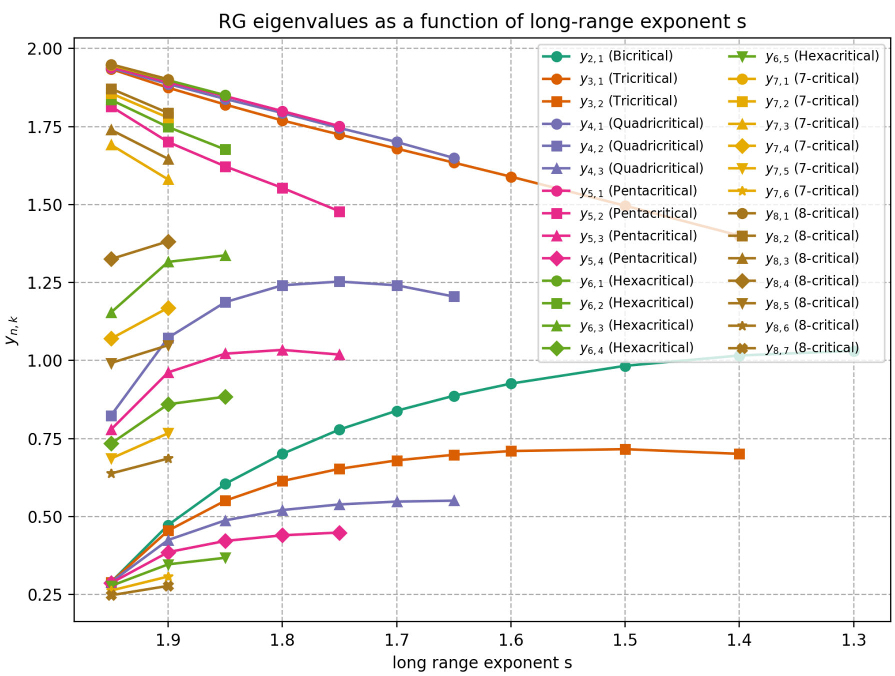

Figure 0.7: The RG eigenvalues for the first eight CFT multicritical models as function of the long-range exponent s.

s factor	1.3	1.4	1.5	1.6	1.65	1.7	1.75	1.8	1.85	1.9	1.95
$y_{2,1}$	1.031	1.016	0.983	0.926	0.887	0.839	0.779	0.701	0.605	0.473	0.290
$y_{3,1}$		1.400	1.496	1.589	1.634	1.679	1.724	1.769	1.820	1.874	1.934
$y_{3,2}$		0.701	0.716	0.710	0.698	0.680	0.653	0.614	0.551	0.455	0.290
$y_{4,1}$					1.649	1.700	1.745	1.793	1.838	1.886	1.937
$y_{4,2}$					1.205	1.241	1.253	1.241	1.187	1.073	0.824
$y_{4,3}$					0.551	0.548	0.539	0.521	0.488	0.425	0.290
$y_{5,1}$							1.751	1.799	1.847	1.892	1.940
$y_{5,2}$							1.478	1.553	1.622	1.700	1.814
$y_{5,3}$							1.019	1.034	1.022	0.962	0.779
$y_{5,4}$							0.449	0.440	0.422	0.386	0.287
$y_{6,1}$									1.850	1.898	1.946
$y_{6,2}$									1.676	1.748	1.835
$y_{6,3}$									1.337	1.316	1.154
$y_{6,4}$									0.884	0.860	0.734
$y_{6,5}$									0.368	0.347	0.278
$y_{7,1}$										1.901	1.946
$y_{7,2}$										1.778	1.856
$y_{7,3}$										1.580	1.691
$y_{7,4}$										1.169	1.070
$y_{7,5}$										0.767	0.686
$y_{7,6}$										0.308	0.263
$y_{8,1}$										1.901	1.949
$y_{8,2}$										1.793	1.871
$y_{8,3}$										1.646	1.739
$y_{8,4}$										1.382	1.325
$y_{8,5}$										1.049	0.992
$y_{8,6}$										0.686	0.638
$y_{8,7}$										0.278	0.248

Table 6: The RG eigenvalues for the first eight CFT multicritical models, as function of the long-range exponent s for a long-ranged model.